RPA PROJECT TITLES

Module 1: Foundations and Skills (5 Projects)

Focus: RPA fundamentals, web technologies, programming, databases, APIs and flowcharts.

1. Automated Student Attendance and Report Generation System

Objective: Read attendance records, calculate attendance percentages and generate reports.

Suggested tools: UiPath, Excel

2. Automated Email Sorting and Attachment Management

Objective: Classify emails, download attachments and organize files into folders.

Suggested tools: UiPath, Outlook

3. Web-Based Product Price Monitoring and Comparison

Objective: Collect permitted public product-price information and create comparison reports.

Suggested tools: UiPath, Power Automate Desktop

4. Automated Employee Onboarding Workflow

Objective: Create employee folders, populate templates and send welcome emails.

Suggested tools: UiPath, Power Automate

5. Database Record Entry and Validation Automation

Objective: Validate spreadsheet records and insert or update database entries.

Suggested tools: UiPath, SQL Server, Python

Module 2: Methodologies and Planning Process (5 Projects)

Focus: Lean, Six Sigma, process selection, implementation planning and ROI.

6. RPA-Based Invoice Processing and Error Reduction

Objective: Validate invoice fields, flag errors and compare processing performance before and after automation.

Suggested tools: UiPath, Excel

7. Automated Leave Request Processing and Approval

Objective: Validate leave balances, route requests and record approval decisions.

Suggested tools: Power Automate, SharePoint

8. RPA Opportunity Identification and ROI Calculator

Objective: Estimate time savings, implementation costs, payback period and ROI for candidate processes.

Suggested tools: Python, Excel

9. Six Sigma-Based Automation of Repetitive Data Entry

Objective: Measure baseline errors, automate data entry and compare before-and-after results.

Suggested tools: UiPath, Excel, Power BI

10. Lean Process Mining and RPA Suitability Assessment

Objective: Analyze process logs, identify bottlenecks and rank processes for automation.

Suggested tools: Python, Celonis or Power BI

Module 3: Vendor Evaluation (5 Projects)

Focus: Platform evaluation, security, monitoring, deployment and RPA Center of Excellence (CoE).

11. Comparative Evaluation of UiPath, Power Automate and Automation Anywhere

Objective: Compare usability, integrations, security, licensing and development effort using a weighted scorecard.

Suggested tools: Excel, Power BI

12. RPA Bot Execution Monitoring and Alert System

Objective: Track bot execution status, failures, duration and alerts through a dashboard.

Suggested tools: UiPath Orchestrator, Power BI

13. Role-Based Access and Audit Log Management for RPA

Objective: Apply access policies, log activities and produce audit reports.

Suggested tools: UiPath Orchestrator, SQL

14. RPA Center of Excellence Dashboard

Objective: Track bots, owners, business value, maintenance status and the automation pipeline.

Suggested tools: Power BI, Excel

15. Secure RPA Credential and Configuration Management

Objective: Use secure credential storage, separate configuration from workflows and audit access.

Suggested tools: UiPath Orchestrator, secure credential store

Module 4: Automation Process Activities (8 Projects)

Focus: Sequences, flowcharts, control flow, variables, collections, arguments, data tables, files and UI controls.

16. Automated Student Mark Processing and Grade Calculation

Objective: Calculate totals, averages, grades and pass/fail status using conditional workflows.

Suggested tools: UiPath, Excel

17. Bulk File Organizer and Document Renaming Bot

Objective: Categorize files by type or date, rename files and generate an activity log.

Suggested tools: UiPath, Python

18. Automated Inventory Reordering System

Objective: Check stock levels, identify reorder requirements and generate purchase requests.

Suggested tools: UiPath, Excel

19. Customer Feedback Classification and Routing

Objective: Classify feedback by category and route records to the appropriate department using rules.

Suggested tools: UiPath, Python

20. Automated Data Entry with Validation and Exception Handling

Objective: Enter form data, validate required fields and record exceptions for review.

Suggested tools: UiPath

21. Multi-Format Data Consolidation and Reporting

Objective: Merge CSV, Excel and text files, remove duplicates and generate summary reports.

Suggested tools: UiPath, Python

22. Automated Expense Claim Verification

Objective: Check expense claims against configured policies and flag unusual or incomplete entries.

Suggested tools: UiPath, Excel

23. Appointment Scheduling and Reminder Automation

Objective: Read appointment data, check scheduling rules and send email reminders.

Suggested tools: Power Automate, Outlook, Excel

Module 5: Advanced Automation Concepts and Techniques (9 Projects)

Focus: Recording, selectors, web automation, scraping, debugging, Excel, PDFs and data extraction.

24. Automated Invoice Data Extraction from PDF Documents

Objective: Extract invoice number, date, vendor and amount; validate the fields and export records.

Suggested tools: UiPath, OCR

25. Web Scraping and Automated Market Research Report

Objective: Extract publicly available information from permitted websites and generate a report.

Suggested tools: UiPath, Python, Beautiful Soup

26. Automated Excel-Based Payroll Preparation

Objective: Calculate salary components, deductions and net salary from input data.

Suggested tools: UiPath, Excel

27. Intelligent Resume Screening and Candidate Shortlisting

Objective: Extract resume details, compare skills with job requirements and rank candidates for human review.

Suggested tools: UiPath, Python, OCR

28. Dynamic Web Form Filling Using Selectors

Objective: Automate web forms using dynamic selectors, validation and exception handling.

Suggested tools: UiPath, Selenium

29. Automated Bank Statement Processing and Expense Categorization

Objective: Extract transaction details, categorize expenses and summarize monthly spending.

Suggested tools: UiPath, OCR, Excel

30. PDF Document Classification and Information Extraction

Objective: Identify document types, extract specified fields and organize documents into folders.

Suggested tools: UiPath Document Understanding

31. Automated Excel Data Cleaning and Duplicate Detection

Objective: Standardize data, remove duplicates, validate records and produce a cleaning report.

Suggested tools: UiPath, Python, Excel

32. Web Portal Automation with Dynamic Controls and Debugging

Objective: Automate a test web portal using selectors, waits, reusable workflows and recovery logic.

Suggested tools: UiPath, Selenium

Module 6: RPA Challenges and RPA in the Enterprise (8 Projects)

Focus: Image and text automation, Citrix, cognitive automation, AI, enterprise integration and business use cases.

33. Intelligent Email-to-Ticket Creation and Routing System

Objective: Read support emails, extract issue details, create tickets and assign categories.

Suggested tools: UiPath, AI services, REST API

34. OCR-Based Automation of Scanned Application Forms

Objective: Read scanned forms, validate extracted data and send uncertain fields for manual verification.

Suggested tools: UiPath, OCR

35. Desktop Application Automation Using Image Recognition

Objective: Identify UI elements visually, interact with desktop controls and handle unexpected screens.

Suggested tools: UiPath, Power Automate Desktop

36. Virtual Desktop and Citrix Data Entry Automation

Objective: Automate a test virtual desktop using keyboard, image and text-based interactions.

Suggested tools: UiPath, Citrix test environment

37. AI-Powered Customer Feedback Sentiment Analysis and Reporting

Objective: Analyze feedback sentiment, summarize trends and generate management reports.

Suggested tools: UiPath, Python, NLP API

38. Automated IT Incident Detection and Notification

Objective: Monitor sample log files, detect configured error patterns and notify support staff.

Suggested tools: Python, UiPath, email

39. Intelligent Purchase Order and Invoice Matching System

Objective: Match purchase orders with invoices, identify discrepancies and prepare exception reports.

Suggested tools: UiPath, OCR, Excel, SQL

40. End-to-End Enterprise Employee Exit Clearance Automation

Objective: Coordinate department clearance, asset return records, account deactivation requests and final status reporting.

Suggested tools: UiPath, Power Automate, SQL, REST API

Recommended Projects for Student Teams

1. Intelligent Invoice Processing (Project 24)

Combines PDF extraction, OCR, data tables, validation, exception handling and Excel reporting. Suitable as a well-rounded demonstration project.

2. Student Attendance Automation (Project 1)

Beginner-friendly project covering sequences, variables, conditional logic, Excel and automated reports.

3. RPA Bot Monitoring Dashboard (Project 12)

Advanced option demonstrating enterprise monitoring, exception reporting and operational metrics.

4. Intelligent Document Processing (Project 30)

Integrates document classification, OCR, data extraction and human verification.